

Telangana – Grade 10 2015 – Mathematics

Maximum Marks:40

Question: If a cylinder and a cone are of the same radius and height, then how many cones full of milk can fill the cylinder? Answer with reasons.

Solution: Let radius be r and height be h .

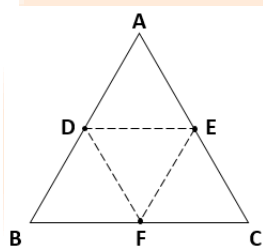
$$\text{Vol of cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Vol of cylinder} = \pi r^2 h$$

On comparison $3 \times \text{cone volume} = \text{vol. of cylinder}$.

Hence, cone needs to be filled 3 times.

Question: In a $\triangle ABC$; D, E and F are the mid-points of AB, AC and BC respectively. If the area of $\triangle DEF$ is 14.4 cm^2 , then find the area of $\triangle ABC$



Solution: Triangles formed by mid-point = $\frac{1}{4} \times \text{Area of full triangle}$.

$$\text{Area of } \triangle ABC = 4 \times \triangle DEF$$

$$= 4 \times 14.4$$

$$= 57.6 \text{ cm}^2$$

Question: When a die is rolled once unbiased, what is the probability of getting a multiple of 3 out of possible outcomes?

Solution: Event = $\{3, 6\}$

$$n(s) = 6$$

$$\text{Probability} = \frac{2}{6} = \frac{1}{3}$$

Question: Show that $\tan^2 \theta - \frac{1}{\cos^2 \theta} = -1$

Solution:

$$\begin{aligned}\tan^2 \theta - \frac{1}{\cos^2 \theta} &= -1 \\ \tan^2 \theta - \sec^2 \theta & \\ &= -(\sec^2 \theta - \tan^2 \theta) \\ &= -1\end{aligned}$$

Question: How many tangents can be drawn to a circle from a point on the same circle? Justify your answer.

Solution: One

Question:

Class Interval	10-25	25-40	40-55	55-70	70-85	85-100
Frequency	2	3	7	6	6	6

How do you find the deviation from the assumed mean for the above data?

Solution:

Class Interval	x	f	fx	$x - \bar{x}$	$(x - \bar{x})^2$
10-25	17.5	2	35	-44.5	1980.25
25-40	32.5	3	97.5	-29.5	870.25
40-55	47.5	7	332.5	-14.5	210.25
55-70	62.5	6	375.0	0.5	0.25
70-85	77.5	6	465.0	15.5	240.25
85-100	92.5	6	555.0	30.5	930.25
Total		30	1860		4231.5

$$\bar{x} = \frac{1860}{30} = 62$$

$$\sigma = \sqrt{\frac{4231.5}{30}} = 11.876$$

Marks:

negative marks:

Subject: Mathematics

Type: para

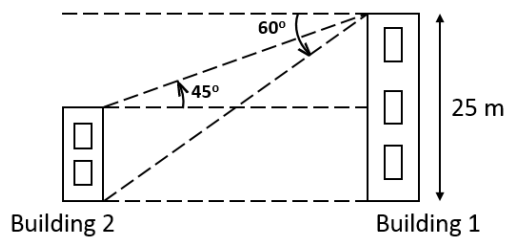
Topics:

Target grade:

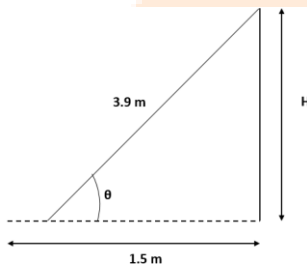
Difficulty:

Question: A person from the top of a building of height 25 m has observed another building's top and bottom at an angle of elevation 45° and at an angle of depression 60° respectively. Draw a diagram for this data.

Solution:



Question: A ladder of 3.9 m length is laid against a wall. The distance between the foot of the wall and the ladder is 1.5 m. Find the height at which the ladder touches the wall.



Solution:

$$\cos \theta = \frac{1.5}{3.9}$$

$$H^2 + 1.5^2 = 3.9^2$$

$$H^2 + 2.25 = 15.21$$

$$H^2 = 15.21 - 2.25$$

$$H = \sqrt{12.96}$$

$$H = 3.6m$$

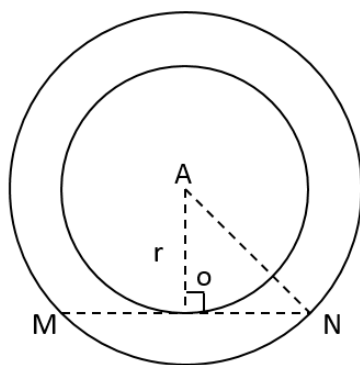
Question: There are 12 red balls, 18 blue balls and 6 white balls in a box. When a ball is drawn at random from the box, what is the probability of not getting a red ball?

Solution: Event of no of Red = {Blue + White balls $18 + 6 = 24$ }

$$P(\bar{R}) = \frac{24}{12+18+6} = \frac{24}{36} = \frac{2}{3}$$

Question: Prove that “in two concentric circles, a chord of the bigger circle, that touches the smaller circle is bisected at the point of contact with the smaller circle.”

Solution:



To prove: $OM = ON$ from fig
 $\angle O = 90^\circ$

In $\triangle AON$

$$ON = \sqrt{AN^2 - AO^2}$$

$$= \sqrt{R^2 - r^2}$$

In $\triangle AOM$

$$OM = \sqrt{AM^2 - AO^2}$$

$$= \sqrt{R^2 - r^2}$$

Hence, MN gets bisected

Question: Show that $(1 + \cot^2 \theta)(1 - \cos \theta)(1 + \cos \theta) = 1$.

Solution:

$$(1 + \cot^2 \theta)(1 - \cos \theta)(1 + \cos \theta)$$

$$\operatorname{cosec}^2 \theta \times (1 - \cos^2 \theta)$$

$$\operatorname{cosec}^2 \theta \times \sin^2 \theta$$

$$= 1$$

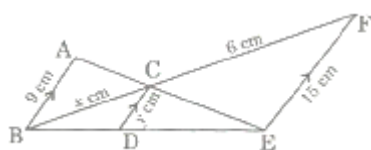
Question: The radius of a spherical balloon increases from 7 cm to 14 cm as air is pumped into it. Find the ratio of the volumes of the balloon before and after pumping the air.

Solution:

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$\frac{\text{Vol. @ 7}}{\text{Vol. @ 14}} = \frac{7^3}{14^3} \times \frac{1}{2^3} = \frac{1}{8}$$

Question: Observe the below diagram and find the values of x and y.



Solution:

$$\frac{AB}{EF} = \frac{BC}{CF} = \frac{9}{15} = \frac{x}{6} = x = 3.6 \text{ cm}$$

$$\frac{CD}{EF} = \frac{x}{x+6} = y = \frac{3.6}{9.6} \times 15 = 5.625 \text{ cm}$$

Question: In a village, an enumerator has surveyed for 25 households. The size of the family (number of family members) and the number of families is tabulated as follows:

Size of the Family (No. of the members)	1-3	3-5	5-7	7-9	9-11
No. of Families	6	7	9	2	2

Find the mode of the data.

OR

There are 100 flash cards labelled from 1 to 100 in a bag. When a card is drawn from the bag at random, what is the probability of getting

- (i) a card with prime number from possible outcomes.
- (ii) a card without prime number from possible outcomes.

Solution:

Modal class = 5 – 7

$$(a) \text{ Mode} = l + h \left(\frac{fm - f1}{2fm - f1 - f2} \right)$$

$$l = 5$$

$$h = 2$$

$$fm = 9$$

$$f1 = 7$$

$$f2 = 2$$

$$\text{Mode} = 5 + 2 \times \left(\frac{9 - 7}{2 \times 9 - 7 - 2} \right)$$

$$= 5 + 2 \times \frac{2}{9} = 5 + \frac{4}{9}$$

$$= \frac{45 + 4}{9} = \frac{49}{9} = 5.\bar{4}$$

(b) Event = {Total primes b/w 1 to 100}

$$= 25$$

$$n(S) = 100$$

$$P = \frac{25}{100} = \frac{1}{4}$$

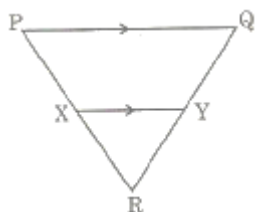
$$\text{Probability of not a prime number} = 1 - \frac{1}{4}$$

$$= \frac{3}{4}$$

Question: Find the value of $\frac{\sec 15^\circ}{\cos 75^\circ} + \frac{\sin 72^\circ}{\cos 18^\circ} - \frac{\tan 33^\circ}{\cot 57^\circ}$

OR

Observe the figure given below:



In $\triangle PQR$, If $XY \parallel PQ$, $\frac{PX}{XR} = \frac{5}{3}$ and $QR = 7.2$ cm, find the length of RY .

Solution:

(a)

$$\begin{aligned} & \frac{\sec 15^\circ}{\cos 75^\circ} + \frac{\sin 72^\circ}{\cos 18^\circ} - \frac{\tan 33^\circ}{\cot 57^\circ} \\ & \frac{\sec(90-15)}{\cos 75} + \frac{\cos(90-72)}{18} + \frac{\cot(90-33)}{57} \\ & \frac{\sec(75)}{\cos(75)} + \frac{\cos(18)}{\cos(18)} - \frac{\cot(57)}{\cot(57)} \\ & 1 + 1 - 1 = 1 \end{aligned}$$

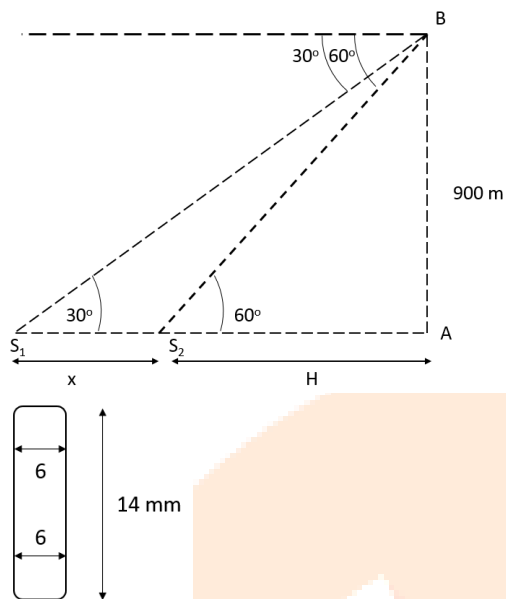
(b) Using B.P.T

$$\begin{aligned} \frac{RX}{XP} &= \frac{RY}{YQ} \\ \frac{3}{5} &= \frac{RY}{7.2 - RY} \\ \Rightarrow 21.6 - 3RY &= 5RY \\ 21.6 &= 8RY \\ RY &= \frac{21.6}{8} = 2.7 \text{ cm} \end{aligned}$$

Question: An observer flying in an aeroplane at an altitude of 900 m observes two ships in front of him, which are in the same direction at an angles of depression of 60° and 30° respectively. Find the distance between the two ships.

OR

A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. If the length of the cylindrical part of the capsule is 14 mm and the diameter of hemisphere is 6 mm, then find the volume of medicine capsule.



Solution: (a)

In $\triangle ABS_2$

$$\tan 60^\circ = \frac{900}{H}$$

$$H = \frac{900}{\sqrt{3}}$$

In $\triangle ABS_1$

$$\tan 30^\circ = \frac{900}{H+x}$$

$$H+x = 900\sqrt{3}$$

$$x = 900\sqrt{3} - \frac{900}{\sqrt{3}}$$

$$= \frac{2700 - 900}{\sqrt{3}} = \frac{1800}{\sqrt{3}}$$

(b)

Volume of Capsule = volume of
Cylinder + vol. of sphere.

$$= \pi \times r^2 h + \frac{4}{3} \pi r^3$$

$$= \frac{22}{7} \times 3 \times 9 \times 14 + \frac{22}{7} \times 3 \times 3 \times 3$$

$$= 396 + 113.14$$

$$= 509.14 \text{ mm}^3$$

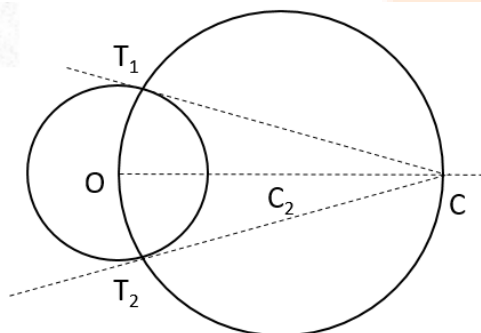
Question: Draw a circle with radius 3 cm and construct a pair of tangents from a point 8 cm away from the centre.

OR

Daily expenditure of 25 householders is given in the following table:

Daily expenditure of a household (in rupees)	100-150	150-200	200-250	250-300	300-350
No. of Families	4	5	12	2	2

Draw a "less than type" cumulative frequency Ogive curve for this data.

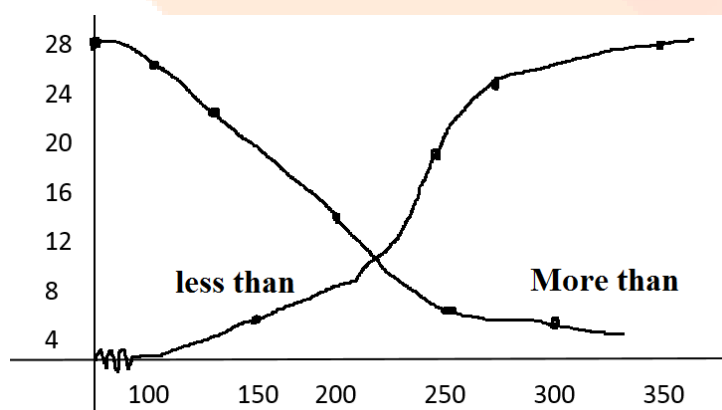


Solution:

(a) Draw circle = 3 cm plot a point 8 cm from O. Draw circle C_2 cutting circle at T_1 and T_2 Join T_1C and T_2C

(b)

Daily	Frequency	Less than	Cf	Greater than	CF
100-150	4	150	4	300	2
150-200	5	200	9	250	4
200-250	12	250	21	200	16
250-300	2	300	23	150	21
300-350	2	350	25	100	25



Question: When we construct a triangle similar to a given triangle as per given scale factor, we construct on the basis of

- (A) SSS similarity.
- (B) AAA similarity.
- (C) Basic proportionality theorem.
- (D) A and C are correct.

Solution: A and C are correct

Marks:

negativemarks:

Subject: Mathematics

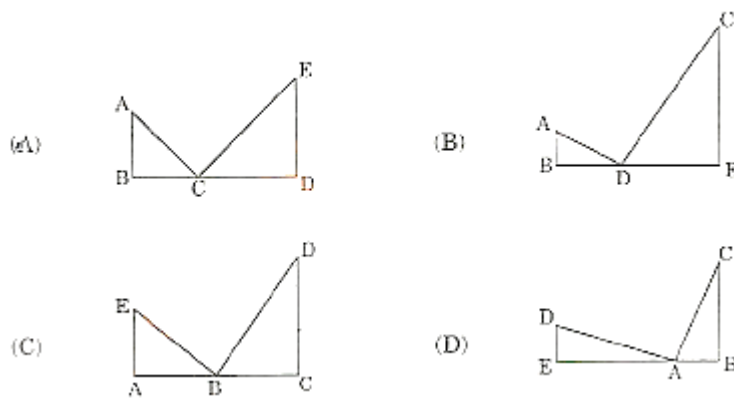
Type: para

Topics:

Targetgrade:

Difficulty:

Question: If $\triangle ABC \sim \triangle EDC$, then which of the following representation of figures is true?



Solution: (a)

Question: The number of pair of tangents can be drawn to a circle, which are parallel to each other, are

- (A) 0
- (B) 2
- (C) 4
- (D) Infinite

Solution: (B)

Question: For a right circular cone with radius = r , height = h and slant height = l , which of the following is not true?

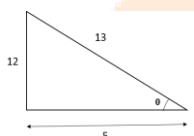
- (A) Always $l > h$
- (B) Always $l > r$
- (C) Always $r > \pi$
- (D) $l^2 = r^2 + h^2$

Solution: (C)

Question: If $\cot A = \frac{5}{12}$, then $\sin A + \cos A$ is

- (A) $\frac{17}{13}$
- (B) $\frac{12}{13}$
- (C) $\frac{5}{13}$
- (D) $\frac{20}{13}$

Solution:



$$\begin{aligned} \sin A + \cos A \\ = \frac{12}{13} + \frac{5}{13} = \frac{17}{13} \end{aligned}$$

Question: Which of the following values is not a possible value of $\sin x$?

- (A) $\frac{3}{4}$
- (B) $\frac{3}{5}$
- (C) $\frac{4}{5}$
- (D) $\frac{5}{4}$

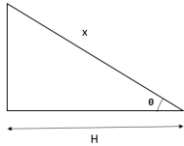
Solution: (D)

$$\frac{5}{4} \text{ because } 0 < \sin x < 1$$

Question: A ladder 'x' meters long is laid against a wall making an angle ' θ ' with the ground. If we want to directly find the distance between the foot of the ladder and the foot of the wall, which trigonometrical ratio should be considered?

- (A) $\sin \theta$
- (B) $\cos \theta$
- (C) $\tan \theta$
- (D) $\cot \theta$

Solution:



$$\frac{H}{x} = \cos \theta$$

$$H = x \cos \theta$$

Question: If $P(E) = 0.82$, then $P(\bar{E}) = \dots$

- (A) 0.18
- (B) 0.28
- (C) 0.38
- (D) $P(E) = P(\bar{E})$

Solution: (A) 0.18 $P(\bar{E}) = 1 - P(E)$
 $= 1 - 0.82$
 $= 0.18$

Question: In “more than Ogive curve”, we consider in drawing

- (A) more than cumulative frequency, lower limits.
- (B) more than cumulative frequency, upper limits.
- (C) less than cumulative frequency, lower limits.
- (D) less than cumulative frequency, upper limits.

Solution: (A)

Question: Observe the following tables:

(1)

Class Interval	Frequency(f)	Class mark(x)	fx

(2)

Class Interval	Frequency(f)	Lower limit (x)	fx

For finding Arithmetic Mean by Direct method, the suggested frequency distribution table is....

- (A) Only (1) is true.
- (B) Only (2) is true.
- (C) (1) and (2) are true.
- (D) None of the above.

Solution: Only (1) is true